

AI Dynamic 3D Camera App & Neo Board

Simple Step-by-Step Guide: App Usage, Features, Camera Ranges & Neo Board Execution

1. How to Use the AI Dynamic Cam App (Step-by-Step)

- Launch the App:** Open `index.html` in Google Chrome or Microsoft Edge, or launch the installed desktop PWA.
- Import 3D Model:** Drag & drop your `.glb` / `.gltf` 3D model into the landing drop zone (optionally attach background audio `.mp3` / `.wav` or camera JSON files). Click **Enter Editor Workspace**.
- Navigate 3D Viewport:**
 - Rotate / Orbit:** Left-click and drag on the 3D viewport canvas.
 - Pan Camera:** Right-click and drag (or `Shift` + Left-click and drag).
 - Zoom In / Out:** Scroll mouse wheel up / down.
 - Fit / Reset View:** Press `F` to fit model to screen; press `R` to reset camera view.
- Add Timeline Keyframes:**
 - Scrub the timeline playhead to any frame (0 to 1000).
 - Position the camera pose in the viewport and press `K` (or click + **Keyframe**).
 - Press `Space` to Play / Pause timeline animation playback.
 - Press `Shift + K` to delete the keyframe at the current playhead frame.

2. Key Features Overview



AI Camera Director

Type natural prompts (e.g. "Hero push in with 35mm lens") or pick motion archetypes (*Hero Push*, *Wide Establish*, *Close Detail*, *Orbit Sweep*, *Spiraling Reveal*).



Lens Focal Length (mm)

Adjust focal length from `1mm` to `120mm`. Quick presets: `24mm` (wide), `35mm` (natural), `50mm` (standard), `85mm` (detail portrait).



Handheld Shake & Audio Sync

Apply procedural shake profiles (*Continuous Handheld*, *Impact Drop*, *Erratic Rumbling*) synced with audio track waveform scrubbing.



Project Export / Import

Export clean keyframe arrays for Three.js/Blender or download full project `.json` files containing all paths, hotspots, and settings.

3. Camera Ranges: Restricting Dynamic Camera Motion

 Core Purpose of Camera Ranges:

If you do **NOT** want dynamic camera motion active across the entire 3D model, you can create specific **Camera Ranges**. Dynamic camera motion will take effect **ONLY inside the added Camera Ranges** in Neo Board. Outside of these added range blocks, the camera stays steady in a fixed static pose. This allows Neo Board text banners, product specifications, and callouts to display cleanly without unwanted camera movement across the full model!

How to Add & Configure Camera Ranges (Step-by-Step):

- 1. Identify the specific frame window where you WANT dynamic camera motion on your model (e.g. Frame 60 to Frame 180 for a 360° product sweep).
- 2. Open the **Camera Ranges** panel in the right drawer inspector.
- 3. Click + **Add Range** to open the inline range editor form.
- 4. Enter your **Start Frame** (e.g. `60`) and **End Frame** (e.g. `180`).
- 5. Configure **Dynamic Tracking Mode**:
 - **Checked ('dynamic: true')**: Dynamic camera motion takes effect **ONLY inside this range** in Neo Board.
 - **Unchecked ('dynamic: false')**: Locks to a fixed static 3D pose, keeping the viewport steady outside dynamic zones.
- 6. Click **Save**. (You can also click directly on empty timeline track lane space to spawn a range block or toggle dynamic mode).

How to Edit an Existing Camera Range:

- **Click Edit on Range Item**: In the Camera Ranges list, click the **Edit** button next to any saved range.
- **Modify Settings**: The form opens in **Edit Range** mode pre-populated with that range's Start Frame, End Frame, Dynamic Tracking checkbox, and Static pose values.
- **Save Changes**: Adjust any parameters and click **Save** to update that specific camera range in real time!

4. What Effect Happens in Neo Board?

Neo Board is the app's integrated presentation overlay HUD system (`.neo-overlay` , `.neo-badge` , `.neo-title` , `.neo-desc` , `.neo-tags`).

Camera Range Mode	3D Viewport Motion	Neo Board Overlay Effect
Inside Added Camera Range (dynamic: true)	Camera actively orbits, dollies, or sweeps along calculated 3D spline curves.	Displays active real-time motion badges (e.g., "DYNAMIC SWEEP") and updates spatial context tags dynamically as the camera sweeps.
Outside Added Camera Range (dynamic: false)	Camera motion is frozen; holds exact static orientation (Yaw, Pitch, Radius, Target XYZ, 0 Shake).	Freezes viewport movement so viewers can comfortably read title cards, specs, and text callouts without camera jitter or motion blur.

Pro Tip: Add Dynamic Camera Ranges only on segments where you need active 3D camera sweeps, and leave static callout segments un-ranged so text stays steady!

5. Keyboard Shortcuts Quick Reference

Key	Action
Space	Play / Pause timeline animation playback
K	Insert keyframe at active playhead frame
Shift + K	Delete keyframe at active playhead frame
Left / Right	Step playhead 1 frame backward / forward
P	Toggle Preview Mode (hides drawer UI for clean presentation)
F	Fit 3D model geometry to viewport center
R	Reset camera view to default orientation